Kosala Gayan

Temporary Lecturer at Department of Mechanical Engineering, University of Moratuwa, Sri Lanka
Kosala Gayan | kosalag@uom.lk | +94770401823

Profile and Research Interests

Highly motivated Mechanical Engineering graduate specialized in Mechatronics at the University of Moratuwa, Sri Lanka, with a strong foundation in Robotics, Machine Learning for computer vision, neural-network-based modeling, Multi-body dynamics, Control systems and currently serve as a graduate instructor for the same university while actively seeking PhD scholarship opportunities. My research interests include Robotics, Autonomous Systems, Industry 4.0, and AI-driven Mechatronic Systems.

Education

Bachelor of the Science of Engineering (Hons.) Aug. 2021 – Jul. 2025
(Washington Accord accredited)
University of Moratuwa *Moratuwa, Sri Lanka*

- Specialization: Mechatronic Systems Engineering Stream
- **GPA:(3.80/4.0) Ranked 8th in Mechanical Engineering (120 students)**

GCE Advanced Level (A/L) Examination Jan. 2017 – Aug. 2019
Sripalee college, Horana *Horana, Sri Lanka*

- Stream: Physical science stream
- Results: **3A Passes, Z Score: 2.3723 (top 1% out of 47,000+ candidates)**

Experience

Academic Experience

Academic instructor Nov. 2025 – May. 2026
University of Moratuwa *Sri Lanka*

Conducting lab classes, tutorials and workshops for undergraduate students and Assisted the senior staff by conducting group discussions, and marking assignments (modules: Instrumentation system design, Industrial automation, Control systems, Mechatronic system design project and MEMS)

Industrial Experience

Trainee Mechatronics Engineer Dec. 2023 – Jun. 2024
Colombo International Container Terminals *Sri Lanka*

Gained practical experience in PLC programming, electrical installations, pneumatic and hydraulic systems, mechanical design, fabrication, and mechanical system maintenance, while developing teamwork and problem solving skills in an industrial environment.

1. **Automated Lubricant Oil Dispensing System Project**
   - Developed a fully automated oil dispensing system using PLC-HMI integration.
   - Implemented QR-coded work orders and barcode-based user verification.
   - Implemented closed-loop dispensing control using solenoid valves with weight-sensor feedback and bidirectional PLC–HMI communication.
2. **QC Inverter Fan Tester Project**
   - Developed a microcontroller-based inverter cooling fan testing device with PWM-based speed controlling.
   - Designed a control algorithm to evaluate fan performance based on encoder feedback and identify faulty units using calibrated speed-response thresholds.

Publications & Patents

[1] A rotor disk in centrifuge machines for the packed cell volume test. In: National Intellectual Property Office of Sri Lanka (NIPO), **(Pending Patent**, Application No: 14245).

[2] **K.A.K. Gayan**, K.D.C.S Kalubowila, M.A.Y. Perera, Y.W.R. Amarasinghe, W.A.D.M. Jayathilaka, Optimal pid controller design for brushed dc motors using cma-es optimization, In: IEEE Engineering Research Express Journal, IOP Publishing, **(Under peer-review)**.

[3] K.D.C.S. Kalubowila, M.A.Y. Perera, **K.A.K. Gayan**, Y.W.R. Amarasinghe, W.A.D.M. Jayathilaka, Design and optimization of a passive isolation architecture for vibration reduction in high-speed packed cell volume centrifuges, In: Journal of Vibration and Control, **(Under peer-review)**.

[4] **K.A.K. Gayan**, Y.W.R. Amarasinghe, W.A.D.M. Jayathilaka, Design & development of vision-based human-following robot with rocker-bogie mobility, In: IEEE Moratuwa Engineering Research Conference (MERCon 2026), **(Accepted)**.

[5] K.D.C.S. Kalubowila, **K.A.K. Gayan**, M.A.Y. Perera, Y.W.R. Amarasinghe, W.A.D.M. Jayathilaka, Passive mechanical retention of capillary tubes for machine-vision compatible high-speed packed cell volume test centrifugation rotor, In: IEEE Moratuwa Engineering Research Conference (MERCon 2026), **(Accepted)**.

[6] K.T.R.J. Weniwelkola, **K.A.K. Gayan**, P.K.V. De Silva, Y.W.R. Amarasinghe, W.A.D.M. Jayathilaka, H.A.G.C. Premachandra, Five-parameter kinematic calibration for low-cost rotational 3d scanners via meta-heuristic optimization, In: IEEE Moratuwa Engineering Research Conference (MERCon 2026), **(Accepted)**.

[7] M.A.Y. Perera, K.D.C.S. Kalubowila, **K.A.K. Gayan**, Y.W.R. Amarasinghe, W.A.D.M. Jayathilaka, Design and manufacturing optimization of a highspeed pcv centrifuge machine, In: Moratuwa Engineering Research Conference (MERCon 2025), DOI: 10.1109/MERCon67903.2025.11217024.

Projects

PORTFOLIO

Design and development of a Novel vision-based PCV analyzer for efficient hematocrit monitoring in dengue patients. Aug. 2021 – Jul. 2025

- Developed a vision-based PCV analyzer that automates manual hematocrit measurement using image processing to calculate red blood cell percentage after centrifugation.
- Implemented High-speed motor controlling using PID to minimize jerks and noise.
- Developed a Passive Isolation Architecture for Vibration Reduction.
- Integrated IoT-enabled data management with a mobile app for real-time transfer of results to medical staff.

Autonomous Miniature Taxi Robot Prototype Mar. 2025 – Jul. 2025

- Developed a PID-tuned control method for precise line following and wall-following navigation.
- Trained a machine learning model on a custom dataset for sign detection.
- Integrated a mobile application for manual/auto mode switching and real-time control.

Design and Development of vision based Human Following Robot. Aug. 2023 – Dec. 2024

- Developed a scaled-down six-wheeled mobile robot including the rocker-bogie suspension and low-height step climbing.
- Perfomed multibody dynamics simulations in MSC Adams to evaluate system forces and motion behavior.

- Developed Media Pipe based human detection and tracking algorithm for real-time autonomous following.
- Built a GUI and web interface offering control and live video feedback, and integrated a mobile app for remote control.

Design and Development of a Fundamental SCADA system Jul. 2024 – Oct. 2024

- Developed an IoT-based SCADA system using an ESP32 for continuous data acquisition, with real-time and historical logs stored in a MySQL database.
- Implemented closed-loop fan speed control by real-time temperature feedback for stable indoor Temperature management.

Design of Mems-Based Hybrid Piezoelectric-Thermoelectric Energy Harvester for Smart Wearable Devices Jul. 2024 – Oct. 2024

- Designed a double S-shaped MEMS structure for vibration energy harvesting and integrated a thermoelectric arrays for thermal energy extraction.

Honours & Awards

Winners of the Undergraduate Inventor of the Year 2025 Competition Jul. 2025

Awarded by Institution of Engineers, Sri Lanka (IESL)

- Achieved 1st place in Final Year Research Project

Dean's List– Faculty of Engineering Jul. 2025

Awarded by University of Moratuwa, Sri Lanka

- Elected due to GPA above 3.80 in 2, 5, 8 semesters.

Skills

Designing(2D, 3D): SolidWorks, AutoCAD
Programming: Python, ROS, MATLAB, C++, C
PLC programming and HMI designing: Siemens, Delta
Application Awareness: MATLAB/Simulink, COMSOL Multiphysics, LabVIEW, FluidSIM, Atmel studio, ANSYS, Adams Multibody Dynamics, Proteus, TIA Portal, Mission Planner
Hardware Skills: Soldering, Electrical Wiring, Microcontrollers, Oscilloscope handling

Extra-Curricular Activities & Volunteer Experience

- **Organizing Committee Member – MERS 2025** Jan. 2026
  Mechanical Engineering Research Symposium (MERS 2025), University of Moratuwa
- **Organizing Committee Member – MechatronX 2023** Jul. 2023
  EXMO 2023, University of Moratuwa
- **Mathematics Tutor – Soyuru Sathkara Seminar Series** May. 2023 – Jun. 2023

References

Prof. Y. W. R. Amarasinghe
Stream Coordinator
Mechatronic Systems Engineering Stream
University of Moratuwa, Sri Lanka
ranama@uom.lk

Dr. W. A. D. M. Jayathilaka
Senior Lecturer
Department of Mechanical Engineering
University of Moratuwa, Sri Lanka
dumithj@uom.lk